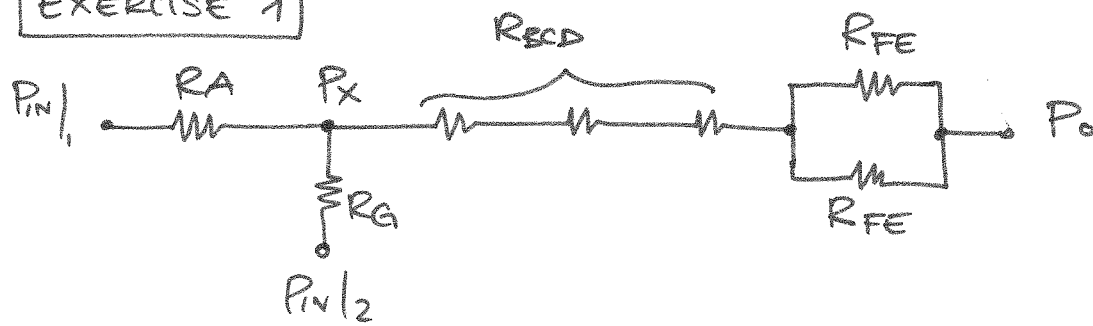
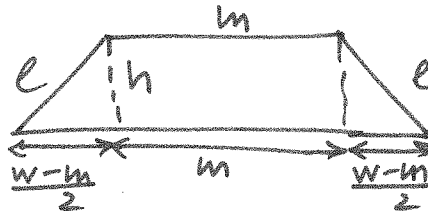


EXERCISE 1



a) $\rightarrow$ Equivalent radius $r_{eq} = \frac{2 \text{ Area}}{\text{perimeter}}$

$$r_{eq} = \frac{2(w+m) \cdot h/2}{w+m+2l}$$



$$l = \sqrt{h^2 + \left(\frac{w-m}{2}\right)^2} = \sqrt{30^2 + \left(\frac{50-30}{2}\right)^2} = 31.6 \mu\text{m}$$

$$r_{eq} = \frac{2(50+30) \cdot \frac{30}{2}}{50+30+2 \cdot 31.6} = 16.75 \rightarrow \boxed{17 \mu\text{m}}$$

$$R = \frac{8\eta \cdot L}{\pi r^4} \quad L_A = 1 \text{ mm} \quad L_{BCD} = 1.7 \text{ mm}$$

$$R_A = \frac{8\eta \cdot L_A}{\pi r^4} = \boxed{3 \cdot 10^{13}}$$

$$R_G = \frac{R_A}{5} = \boxed{6 \cdot 10^{12}}$$

$$R_{BCD} = R_A \cdot \frac{1.7 \text{ mm}}{1 \text{ mm}} = \boxed{5.18 \cdot 10^{13}}$$

$$r_{eq|EF} = \frac{2\left(\frac{w}{2} + \frac{m}{2}\right) \cdot h/2}{\frac{w}{2} + \frac{m}{2} + 2\sqrt{h^2 + \left(\frac{w/2 - m/2}{2}\right)^2}} = \boxed{12 \mu\text{m}}$$

$$R_{EF} = \frac{8\eta L_{EF}}{\pi r_{eq|EF}^4} = \boxed{8.6 \cdot 10^{13}}$$

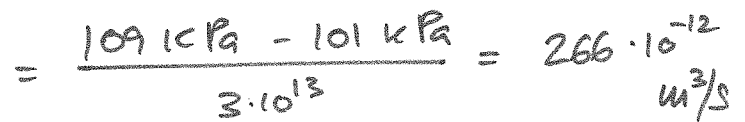
$$Q|_c = \frac{1 \text{ mm}}{1 \text{ mm}} \cdot \text{Area} = \frac{1 \text{ mm}}{1 \text{ mm}} (w+m) \cdot \frac{h}{2} = 20 \cdot 10^{-15} \text{ m}^3/\text{s}$$

$$P_x = P_o + \Delta P = P_o + Q \left(R_{BCD} + \frac{R_{FE}}{2} \right) =$$

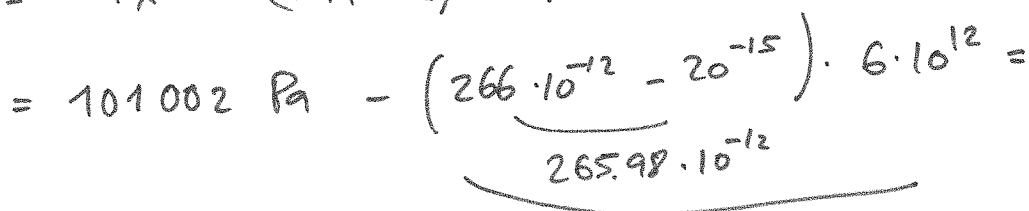
$$= 101 \text{ kPa} + 20 \cdot 10^{-15} \left(5.18 \cdot 10^{13} + \frac{8.6}{2} 10^{13} \right) \approx P_o$$

$\underline{101,002 \text{ Pa}}$

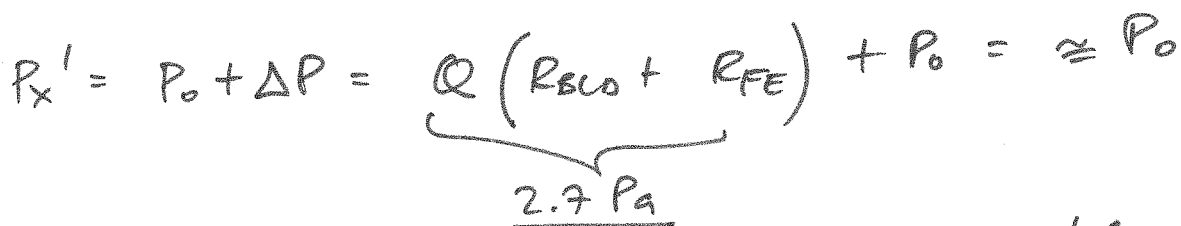
2 Pa (Extremely low)



$$Q_G < 0 \rightarrow$$

$$(P_{n2} < P_0)$$


1600 Pa

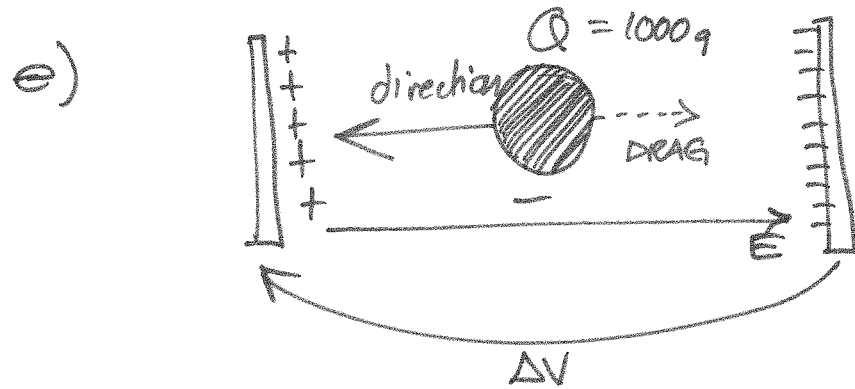


c) PBS cannot enter since $P_{in}/2$ is in aspiration cross-section C only filled with water

$$d) \quad v|_{EOWM} = \frac{\epsilon \cdot E \cdot \xi}{\eta} = 1 \text{ mm/min}$$

$$E = \frac{V \cdot \eta}{\epsilon \cdot \xi} = \frac{\frac{10^{-3}}{60} \cdot 10^{-3}}{\frac{80}{\epsilon_r} \cdot \frac{8.86 \cdot 10^{-12}}{\epsilon_0} \cdot 50 \cdot 10^{-3}} = 470 \text{ V/m}$$

$$\Delta V = E \cdot C = 470 \text{ V/m} \cdot 1 \text{ mm} = 0.47 \sim \boxed{0.5 \text{ V}}$$



MOBILITY

$$\mu = \frac{Q}{6\pi r \cdot \eta}$$

velocity $v_d = \mu \cdot E$

$$E = \frac{10 \text{ V}}{1 \text{ mm}}$$

$$v_d = \frac{Q}{6\pi \cdot r \cdot \eta} \cdot \frac{10 \text{ V}}{1 \text{ mm}} = \frac{1000 \cdot 1.6 \cdot 10^{-19}}{6\pi \cdot 1 \mu\text{m} \cdot \eta} \cdot \frac{10}{1 \text{ mm}} = 2.848 \cdot 10^{-6} \text{ m/s}$$

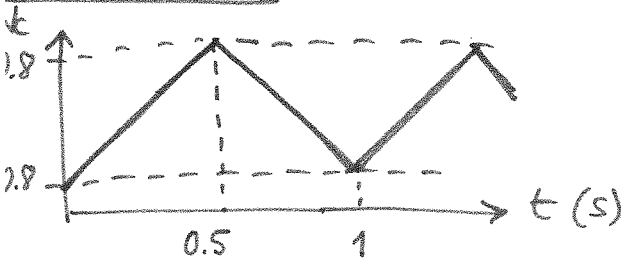
$$\sim \boxed{17 \mu\text{m/sec}}$$

17 $\mu\text{m/sec}$

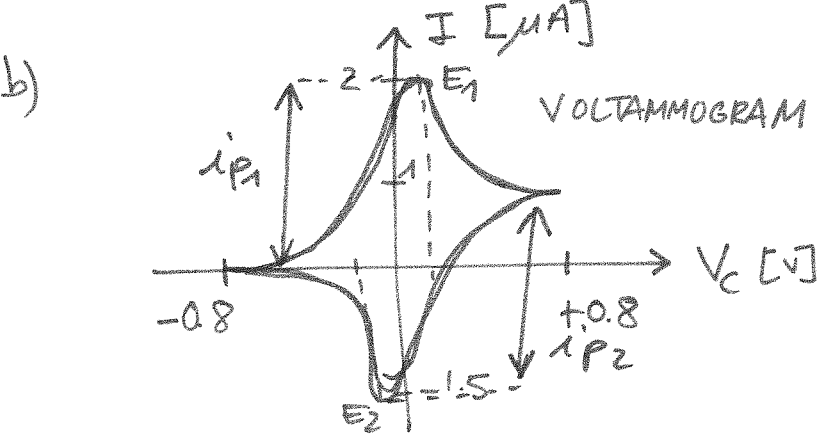
$$\sim \boxed{1} \text{ mm/min}$$

(1 mm/min)

EXERCISE 2



a) $SR = \frac{dV}{dt} = \frac{0.8 + 0.8}{0.5} = 3.2 \text{ V/s}$



c) from the voltammogram, :

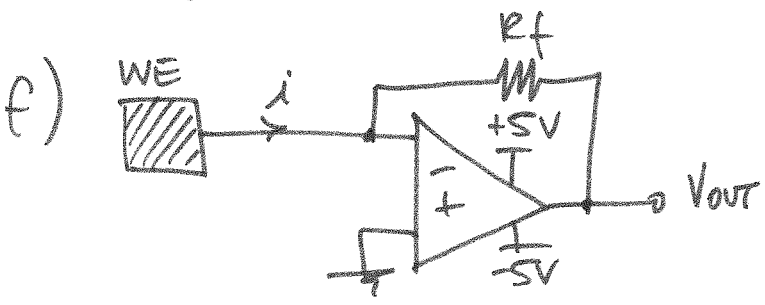
$E_1 \sim 0.12 \text{ V}$
 $E_2 \sim -0.12 \text{ V}$

$\Delta E = E_1 - E_2 \sim 240 \text{ mV} > 59 \text{ mV}$ (ideal separation), but since $i_{p1} \cong i_{p2}$ the reaction is reversible.
 ($\sim 2 \mu\text{A}$)

d) $BW = 40 \cdot SR \Rightarrow 40 \cdot 3.2 = 128 \text{ Hz}$ min. bandwidth

e) $\Delta E|_{\text{ideal}} = 59 \text{ mV}$ $E_{\text{max}} = 10\% \cdot \Delta E = 5.9 \text{ mV}$

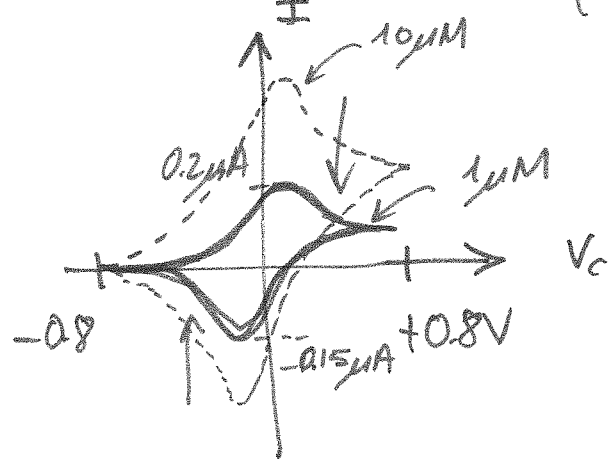
Accuracy better than 5.9 mV



$i_{\text{max}} = 2 \mu\text{A}$
 $i_{\text{min}} = -1.5 \mu\text{A}$

$Rf|_{\text{max}} = \frac{5 \text{ V}}{2 \mu\text{A}} = 2.5 \text{ M}\Omega$

g) Linear scaling of the current by a factor 10:
 (with the concentration)



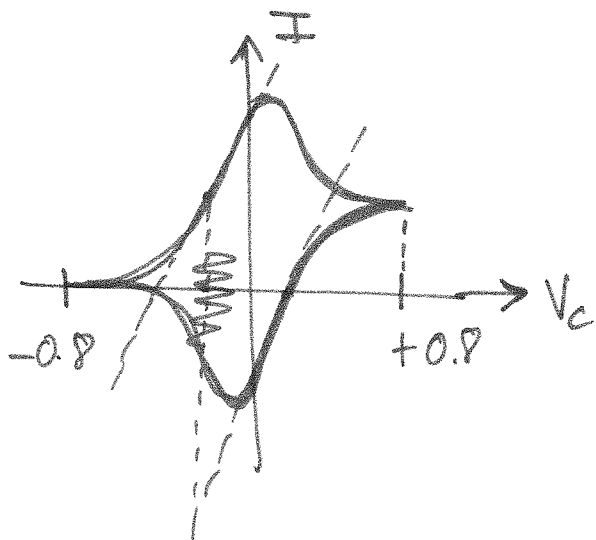
h) $R_{CT} = \left(\frac{\partial i}{\partial V_c} \right)^{-1}$

$R_{CT} \text{ min} \rightarrow$ highest slope of the voltammogram

from the plot

$\frac{\partial i}{\partial V_c} / \text{max}$ in

$V_c = -0.1 \text{ V}$
 and
 $V_c \sim 0 \text{ V}$



i) CMOS: ELECTRODES ISSUES

